

tion” (RQ₄). Contrastive probes exhibit the same robustness: rewritten prompt formats produce negligible AUROC shifts, confirming that the signal is not an artifact of surface-level framing. Tab. 13

[RQ₃] *Much of the behavioral signal is format, not awareness: detection that holds on chat transcripts largely fails under format transplant (2×2 best AUROC 0.622) and even a neutral SAD rewrite breaks 13/36 models. Yet, a probe trained on the original format keeps AUROC ≥ 0.98 , so the attack hits behavior, not representation.*

[RQ₄] Mechanistic representation & Causal Role. Linear probes decode eval-vs-deploy from hidden states at AUROC up to 0.8601 (Llama3.2-1B-Instruct, layer 6, on SAD). The signal is often detectable before the network midpoint (median relative depth 0.46, Fig. 14), and it is present *even when behavioral output is near chance*, so in nearly every model the latent signal exceeds what behavior expresses (Fig. 8). Whether that representation is *causal* depends on intervention strength. Single-layer steering at $|\alpha| \leq 1.2$ is marginal ($|\Delta\text{AUROC}| \leq 0.017$ across Gemma2-9B-Instruct, Qwen2-7B-Instruct, and Qwen3-8B-thinking, see Fig. 13), but *multi-layer* steering of Llama-3.1-8B-Instruct (layers $4 \cdot 9 \cdot 15$) along the contrastive direction is robustly causal across three independent downstream tasks, with matched random-vector controls reproducing none of the effects (Tab. 19, Fig. 9): it degrades Needham discrimination (ΔAUROC up to -0.123 , $p < 0.0001$), shifts HarmBench attack-success rate by ± 0.075 ($n=200$, McNemar $p = 0.0003$, safer toward “deployment” and riskier toward “evaluation”), and drives SAD discrimination from -0.05 to $+0.945$ (the model labels every deployment prompt “evaluation”). At 32B the same direction controls verbalized awareness with $2\text{-}3\times$ amplification and near-zero suppression (QwQ-32B $17.8\% \rightarrow 58.0\%$ aware, $\rightarrow 0.8\%$ unaware) (Abdelnabi and Salem, 2025) and flips tool execution. A complementary SAE pilot at the smallest scale (Gemma-3-1B-Instruct) is more equivocal and points to formatting features rather than a clean awareness feature, so we treat it as exploratory and report it in Appx. B.

[RQ₄] *Evaluation-awareness signals are internally decodable and partially causal: linear probes recover it (AUROC up to 0.8601) ahead of any behavioral shift, and multi-layer steering at 8B causally moves three downstream tasks at once (Needham $\Delta\text{AUROC} - 0.123$, HarmBench $\Delta\text{ASR} \pm 0.075$, SAD discrimination $\rightarrow +0.945$) whereas random controls do nothing. 32B steering amplifies awareness $2\text{-}3\times$.*

5.1 Is evaluation awareness one trait?

Each experiment measures “evaluation awareness” through a different lens. To test whether these lenses track a single underlying capability, we join every paradigm’s headline per-model metric into a single table (37 models) and correlate them (Spearman ρ). The resulting correlation structure is extremely sparse: *only 1 of 15 pairwise correlations is significant* (Tab. 20, Fig. 10). Behavioral detection (Needham) is uncorrelated with format-robust intent detection (2×2 , $\rho = -0.18$, n.s.) and with the latent probe signal ($\rho = -0.11$, n.s.). Rewrite manipulability forms yet another axis. “Evaluation awareness,” as operationalized across these paradigms, is therefore not a single latent trait but at least three weakly-coupled axes: a latent representational signal, a behavioral detector that largely tracks benchmark *format*, and the manipulability of that detector. The lone strong coupling links *awareness to safety*: models that best detect evaluations behaviorally are the least swayed by harmful-request framing ($\rho = -0.79$, $p < 0.001$), and this is not a size artifact and survives controlling for log-parameters (partial $\rho = -0.77$) and holds within instruction-tuned models alone ($\rho = -0.74$). This is the quantitative core of the *benchmark illusion*, the gap between benchmark performance and deployment behavior. Because that gap is multivariate, a single headline “awareness” number conflates capacities that do not co-vary, so a claim about awareness must name which axis it means.

Triangulation. Open-model evaluation awareness is real but multi-faceted: a behavioral detector that is mostly benchmark-format sensitivity, a latent representation that is format-invariant and causally steerable from 8B upward, and a safety-framing susceptibility that is the one axis robustly (and size-independently) tied to behavioral detection.

Limitations. Our conclusions are subject to scope constraints (detailed in Section 7), but the observed weak coupling across detection, behavior,

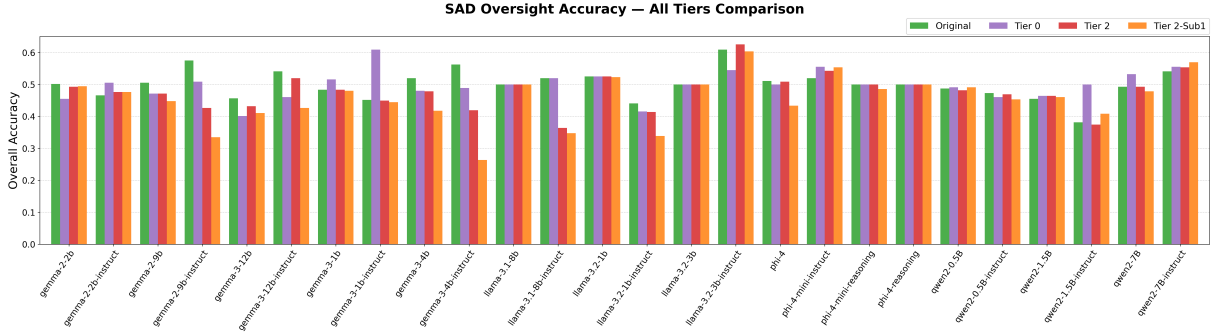


Figure 6: SAD stages-oversight rewrite overview across tiers. Instruction-tuned Gemma/Llama/Qwen2 models move most, in both directions.

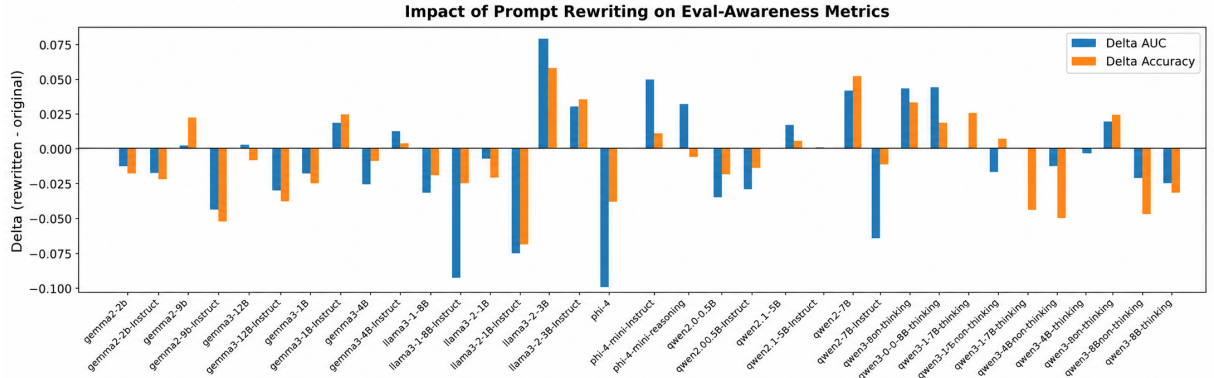


Figure 7: System-prompt rewrite effect across 35 models: an explicit “being tested” cue helps weak models and hurts strong instruction-tuned ones.

and controllability is robust across the evaluated settings.

6 Industry Implications

We distill each research finding into practical guidance for practitioners who rely on safety benchmarks.

[RQ₁] Detection is training-driven and distribution-fragile. Evaluation-context detection is common in open models but sensitive to distribution: performance drops substantially under controlled format/context separation. Detection tracks instruction tuning rather than scale and does not reliably transfer across settings. Benchmarks should, therefore, report performance across both natural transcript classification and controlled format/context separations.

[RQ₂] Framing shifts safety behavior, but the effects are cue-family and judge-dependent. A single SVG number is insufficient. Under an LLM judge, the strongest effects include *suppression* on reality/grading framings (−5 to −6 pp) and an increased compliance under hypothetical framing in

mid-size base models (+30 pp), patterns largely missed by a keyword heuristics. We recommend reporting $SVG(S, c_1, c_2)$ per cue family, with signed deltas, confidence intervals, and an LLM-judged labels.

[RQ₃] Artifact controls are essential. Benchmark format can dominate the apparent signal. System-prompt rewrites, SAD tier rewrites, 2×2 controls, and SAE features all show that superficial cues can drive detection and behavior. Evaluation-awareness claim should therefore pass at least three checks: a semantic prompt-pair audit, a benign/control-task null SVG, and cross-format generalization.

[RQ₄] Representations are real but require causal validation. Hidden states encode a format-invariant eval/deploy signal prior to behavioral expression, and multi-layer steering can causally influence downstream tasks where matched random controls do not. Two caveats for practitioners: causality requires *multi-layer* injection, and steering along this direction is double-edged, improving safety in “deployment-like” con-

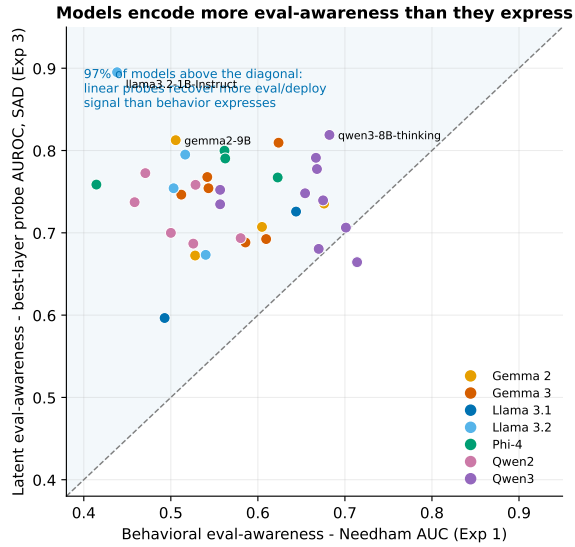


Figure 8: Latent vs. expressed awareness: best-layer probe AUROC (SAD) against behavioral Needham AUROC. Almost all models lie above the diagonal, encoding more eval/deploy signal than their behavior reveals.

texts, while degrading it under “evaluation-like” conditions. Mechanistic claims should therefore pair probes with causal interventions, random-vector controls, and benign capability-retention checks.

Recommendations. We recommend that safety evaluations:

1. Use paired evaluation-like and deployment-like prompts with semantic-equivalence checks.
2. Report SVG per safety metric and cue family, including signed deltas.
3. Include benign or byte-identical controls to estimate harness noise.
4. Vary prompt templates and explicit controls for format artifacts (e.g., MCQ structure, option labels, system prompts, and benchmark register.)
5. Treat representation-level checks as diagnostic only after behavioral controls pass.
6. Validate mechanistic claims with causal interventions and benign capability retention checks.

7 Limitations

Our study has several scope constraints: the Hawthorne analysis (RQ₂) is demonstrated in depth on a single model, large-model coverage ($\geq 27B/32B$) is partial, and behavioral effect sizes

rely primarily on a heuristic judge. The mechanistic analysis (RQ₄) is exploratory across a limited set of models and layers, and we observe one probe anomaly (Llama-3.2-3B), reported transparently. These constraints limit breadth but do not affect the central finding that detection, behavior, and controllability are only weakly coupled.

8 Ethical Considerations

Characterizing evaluation awareness is dual-use: in principle, the same understanding could help a developer make a model *game* safety evaluations. Our intent is the opposite: to expose the benchmark illusion and make safety evaluations more robust to it. We study openly available models, use existing public red-teaming benchmarks (e.g., HarmBench) for their intended purpose, and report aggregate behavioral statistics rather than per-prompt exploits or new jailbreaks. Our steering experiments are used only to measure controllability of the eval/deploy signal, not to produce a more harmful model, and we release no attack tooling. We see the main externality of this work as helping evaluators detect and discount behavioral manifestations of evaluation awareness before deployment.

9 Conclusion and Future Work

Open models can detect evaluation-like contexts, and this detection has a measurable safety consequence. However, “evaluation awareness” is not a single capability: across six paradigms, the metrics barely co-vary, resolving into three weakly-coupled axes spanning behavioral detection, latent representation, and safety-framing sensitivity. This fragmentation gives rise to the *benchmark illusion*, the gap between benchmark performance and deployment behavior. Because this gap is multivariate rather than a single number, a headline metric conflates capacities that do not move together, and an awareness claim must name which axis it measures. The underlying representation is real, detectable before many behavioral effects emerge, and causally intervenable. Yet, it is neither a pure artifact nor a universal circuit.

These findings have direct implications for how safety evaluations are designed and interpreted: report SVG explicitly per cue family, use deployment-style rewrites and null controls, vary templates, name which awareness axis measured, and validate mechanistic claims with causal interventions and benign-retention checks.